

Lab 3 Profiling Report - Abrar

Sorting Algorithm Comparisons:

- Selection sort's nested loops make it so that every remaining number in the list has to be scanned and compared regardless of their order. This results in it having a $O(n^2)$ complexity (134,225,920 comparisons for 16,384 items, taking 6.458991 seconds)
- Quicksort uses a recursive partitioning technique, reducing the complexity significantly to a $O(n \log n)$. On the same list of 16,384 items, Quicksort needed only 265,409 comparisons, finishing in 0.031934 seconds, roughly 200 times faster than Selection Sort.

Fibonacci Sequence Comparisons:

- The naive recursive fibonacci function keeps redoing subcalculations, which results in an exponential $O(2^n)$ complexity, requiring 1,664,079 calls to calculate $n = 30$.
- Using the memoization technique to write a more efficient function reduces the complexity to a linear $O(n)$. This resulted in $n = 30$ being calculated in only 57 calls.